Adaptivity and Optimism

An Improved Exponentiated Gradient Algorithm

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Outline

Algorithm: Hedge — expert advice protocol

1: Initialize weights $w_{1,i} = 1$ for all i.

2: For $t = 1, \cdots, T$, form $p_{t,i} = \frac{w_{t,i}}{\sum_{j=1}^{n} w_{t,j}}$ learner plays p_t.

3: Loss vector $z_t \in [-1, 1]^n$ is revealed; learner suffers $p_t^\top z_t$.

4: For each i, update $w_{t+1,i} = w_{t,i} \exp\left(-\eta z_{t,i}\right)$.

Regret: for $u \in \Delta_n$, $R_{T(u)} := \sum_{t=1}^{T} (p_t - u)^\top z_t$.

Standard upper bound.

$$R_{T(u)} \leq \frac{\log n}{\eta} + \eta \sum_{t=1}^{T} \|z_t\|_\infty^2$$

Algorithm: MW2 — adaptive weights

1: Initialize weights $w_{1,i} = 1$ for all i.

2: For $t = 1, \cdots, T$, form $p_{t,i} = \frac{w_{t,i}}{\sum_{j=1}^{n} w_{t,j}}$ learner plays p_t.

3: Loss vector $z_t \in [-1, 1]^n$ is revealed; learner suffers $p_t^\top z_t$.

4: For each i, update $w_{t+1,i} = w_{t,i} \left(1 - \eta z_{t,i}\right)$.

Bound:

$$\mathrm{Regret}(u) \leq \frac{\log n}{\eta} + \eta \sum_i u_i \sum_t z_{t,i}^2$$

The second term is now averaged by the comparator u.

Hedge pays for the worst coordinate on every round:

$$\eta \sum_t \|z_t\|_\infty^2$$

MW2 pays only for the comparator-weighted coordinates:

$$\eta \sum_i u_i \sum_t z_{t,i}^2$$

For the best expert $u = e_{i^\star}$, this becomes

$$\eta \sum_t z_{t,i^\star}^2$$

Irrelevant noisy experts no longer enlarge the second term.

Algorithm: Optimistic EG – hints

1: Initialize scores $\theta_{1,i} = 0$ for all i.

2: For $t = 1, \cdots, T$, using hint m_t, form $p_{t,i} \propto \exp\left(\theta_{t,i} - \eta m_{t,i}\right)$ learner plays p_t.

3: Loss vector $z_t \in [-1, 1]^n$ is revealed; learner suffers $p_t^\top z_t$.

4: For each i, update $\theta_{t+1,i} = \theta_{t,i} - \eta z_{t,i}$.

Bound:

$$R_{T(u)} \leq \frac{\log n}{\eta} + \eta \sum_t \|z_t - m_t\|_\infty^2$$

The second term now measures hint error. With $m_t = z_{t-1}$, this becomes a path-length quantity.

1.5 Two Ways to Improve the Second Term

We now have two independent changes to the Hedge template:

Adaptive weights

Change the weight update.

Worst-coordinate scale becomes comparator scale:

$$\sum_i u_i \sum_t z_{t,i}^2.$$

Optimism / hints

Change the probability before observing z_t.

Loss scale becomes hint-error scale:

$$\sum_t \|z_t - m_t\|_\infty^2.$$

The paper combines both changes in one algorithm.

1.6 Why Improve the Second Term?

Many refinements keep the first term $\frac{\log n}{\eta}$ and replace the second term by a more local quantity.

- **Standard EG:** $\sum_{t=1}^{T} \|z_t\|_\infty^2$.
- **Adaptive / MW2:** $\sum_{i=1}^{n} u_i \sum_{t=1}^{T} z_{t,i}^2$.
- **Optimistic methods:** $\sum_{t=1}^{T} \|z_t - m_t\|_\infty^2$.
- **This paper:** $\sum_{i=1}^{n} u_i \sum_{t=1}^{T} \left(z_{t,i} - m_{t,i}\right)^2$.

Drawback of $\|z_t\|_\infty^2$: it uses the worst coordinate at each round, not the comparator chosen in hindsight.

A noisy irrelevant expert can enlarge the bound even when the best expert is stable or predictable.

The paper's key move:

adaptive best-expert scale + optimistic hint error

Desired second term:

$$\sum_t \left(z_{t,i^\star} - m_{t,i^\star}\right)^2.$$

This yields one algorithm whose guarantee becomes variance or path length after choosing the hint sequence.

Algorithm: Adaptive Optimistic EG — adaptive weights + hints

1: Initialize $\beta_{1,i} = 0$.

2: For $t = 1, \cdots, T$, using hint m_t, form $p_{t,i} \propto \exp\left(\beta_{t,i} - \eta m_{t,i}\right)$ learner plays p_t.

3: Loss vector $z_t \in [-1, 1]^n$ is revealed; learner suffers $p_t^\top z_t$.

4: For each i, update $\beta_{t+1,i} = \beta_{t,i} - \eta z_{t,i} - \eta^2 \left(z_{t,i} - m_{t,i}\right)^2$.

The red correction downweights experts whose hints were inaccurate.

Theorem (Adaptive Exponentiated Gradient)

If $\|z_t\|_\infty \le 1$, $\|m_t\|_\infty \le 1$, and $0 < \eta \le \frac{1}{4}$, then for every $u \in \Delta_n$,

$$\mathrm{Regret}(u) \le \frac{\log n}{\eta} + \eta \sum_i u_i \sum_t \left(z_{t,i} - m_{t,i}\right)^2.$$

This is exactly the desired combination: the comparator chooses the relevant coordinates, and the hint error chooses the relevant scale.

3.3 Path-Length Result

Set $m_t = z_{t-1}$ and take $u = e_{i^\star}$:

$$\mathrm{Regret}(i^\star) \le \frac{\log n}{\eta} + \eta D_{i^\star}, \quad D_i := \sum_t \big(z_{t,i} - z_{t-1,i}\big)^2.$$

This answers Kale's question: can the path-length bound depend on the best expert rather than the worst coordinate?

The same upper bound has different interpretations:

Hint	**Error term**	**Regret scale**
$m_t = 0$	z_t	$S_{i^\star}$
$m_t = \frac{1}{t}\sum_{s<t} z_s$	$z_t - m_t$	$V_{i^\star}$
$m_t = z_{t-1}$	$z_t - z_{t-1}$	$D_{i^\star}$

The paper uses this to unify second-moment, variance, and path-length bounds.

- $S_{i^\star}$ is small when the best expert's losses are close to zero.
- $V_{i^\star}$ is small when the best expert's losses are nearly constant.
- $D_{i^\star}$ is small when the best expert's losses move slowly.

The path-length view is strongest when the best expert may have large losses, but those losses are predictable from the previous round.

4.3 Comparison of Bounds

Algorithm	**Scale in bound**	**What it misses**
EG / Hedge	S_∞	best expert may be easy
Cesa-Bianchi et al.	$S_{i^\star}$	no variance/path adaptivity
Hazan–Kale	$\max_i V_i$	depends on all experts
Chiang et al.	D_∞	depends on worst movement
AEG-Path	$D_{i^\star}$	best expert only

Even if $D_{i^\star} = \Theta(1)$, the other scales can be $\Theta(T)$.

5.1 Proof Tool: Log-Sum-Exp Potential

Use the normalizing potential

$$L(\beta) := \log\left(\sum_i \exp(\beta_i)\right).$$

The weights are just the normalized exponential scores:

$$w_{i(\beta)} = \frac{\exp(\beta_i)}{\sum_j \exp(\beta_j)}.$$

So the proof can be read as a statement about how $L(\beta)$ changes under exponentiated updates.

The proof asks the correction to make

$$L(\beta_t - \eta z_t - \eta^2 a_t) \le L(\beta_t - \eta m_t) - \eta w_t^\top (z_t - m_t).$$

With $a_{t,i} = \left(z_{t,i} - m_{t,i}\right)^2$, the scalar input is

$$\exp(-x - x^2) \le 1 - x \quad \left(|x| \le \frac{1}{2}\right).$$

Then the potential differences telescope into the stated regret bound.

The paper writes the general version in mirror-descent language, using a Fenchel-conjugate potential.

Replace distributions by density matrices:

$$W_t \in \mathcal{S}_+^n, \quad \mathrm{tr}(W_t) = 1, \quad \mathrm{loss} = \mathrm{tr}(W_t Z_t).$$

The analogous update is

$$B_{t+1} = B_t - \eta Z_t - \eta^2 (Z_t - M_t)^2, \quad W_t = \frac{\exp(B_t - \eta M_t)}{\mathrm{tr}(\exp(B_t - \eta M_t))}.$$

The noncommutative step uses Golden–Thompson:

$$\mathrm{tr}(\exp(A + B)) \leq \mathrm{tr}(\exp(A)\exp(B)).$$

5.4 Backup: Optimizing the Path Bound

From

$$\mathrm{Regret}(i^\star) \le \frac{\log n}{\eta} + \eta D_{i^\star},$$

the best fixed step size is

$$\eta = \sqrt{\frac{\log n}{D_{i^\star}}}.$$

When this satisfies $\eta \le \frac{1}{4}$,

$$\mathrm{Regret}(i^\star) \le 2\sqrt{D_{i^\star} \log n}.$$

If $D_{i^\star}$ is unknown, adaptive tuning is possible but the paper's clean bound becomes weaker.